Solution: Deviation from assumed mean

Sl. No	(X)	d=X-A (A=57)	d^2
1	43	-14	196
2	48	-9	81
3	65	8	64
4	57	0	0
5	31	-26	676
6	60	3	9
7	37	-20	400
8	48	-9	81
9	78	21	441
10	59	2	4
N=10		$\Sigma d = -44$	$\Sigma d^2=1952$

$$\sigma = \sqrt{\frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2} = \sqrt{\frac{1952}{10} - \left(\frac{-44}{10}\right)^2}$$

$$= \sqrt{195.2\text{-}19.36} = \sqrt{175.84} = 13.26,$$

Answer = 13.26

12.12 Correlation (ϒ)

Introduction

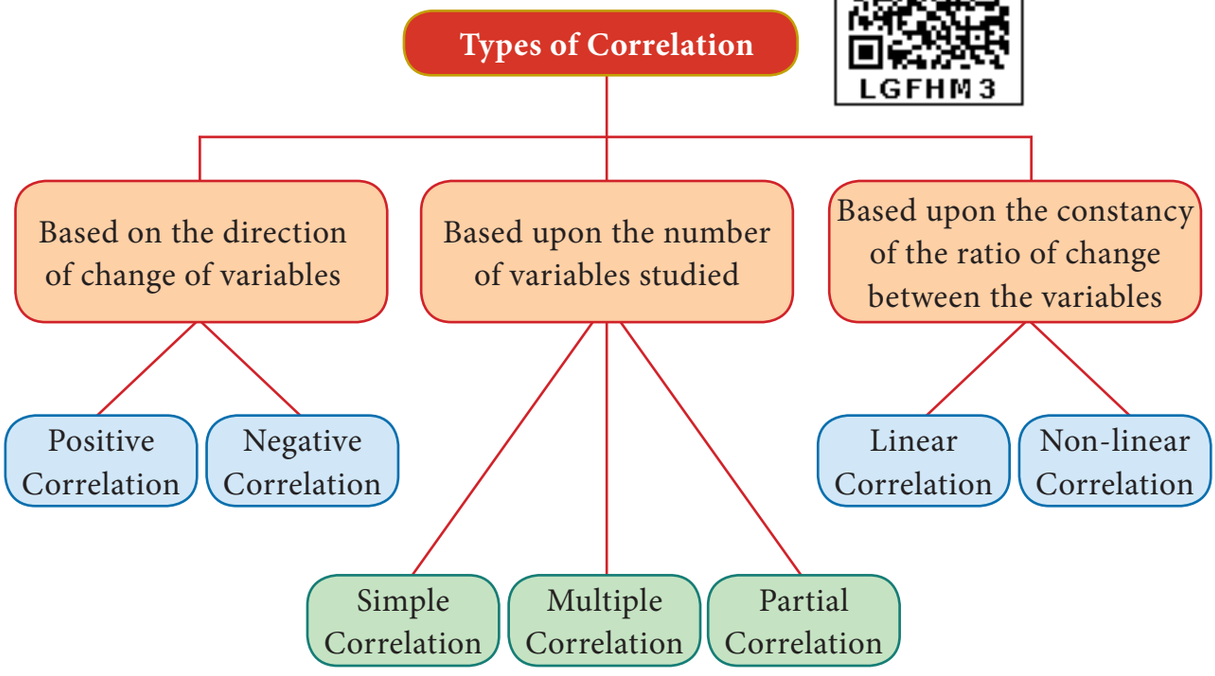
Correlation is a statistical device that helps to analyse the covariation of two or more variables. Sir Francis Galton, is responsible for the calculation of correlation coefficient.

Types of Correlation


Karl Pearson

Correlation is classified in several different ways. Three of the most important ways of classifying correlation are:

LGFHM3



Type I: Based on the direction of change of variables

Correlation is classified into two types as Positive correlation and Negative Correlation based on the direction of change of the variables.

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